AISC5000: Homework 2

Total: 100 points

Question 1.

A disease occurs in 2% of a population. A diagnostic test satisfies

$$P(T = 1 \mid C = 1) = 0.95, \qquad P(T = 1 \mid C = 0) = 0.04,$$

where $C = 1$ denotes disease and $T = 1$ denotes a positive test.

(a) Write down

$$P(C = 1), \qquad P(C = 0),$$

and all four conditional probabilities $P(T = t \mid C = c)$.

(b) Using the sum and product rules, calculate

$$P(T = 1).$$

(c) Using Bayes' theorem, calculate

$$P(C = 1 \mid T = 1).$$

(d) Calculate

$$P(C = 1 \mid T = 0).$$

(e) Explain why

$$P(C = 1 \mid T = 1) \neq P(T = 1 \mid C = 1).$$

In your explanation, identify the roles of the *prior*, *likelihood*, and *posterior*.

(f) Suppose instead that T were independent of C. Show that

$$P(C \mid T) = P(C),$$

and explain why such a test would provide no information about disease status.

Question 2.

Let X have density

$$p_X(x) = \begin{cases} 2x, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Part I

(a) Verify that $p_X(x)$ is a valid probability density.

(b) Calculate

$$P\left(X < \frac{1}{2}\right).$$

(c) Calculate

$$\mathbb{E}[X] \qquad \text{and} \qquad \text{Var}(X).$$

Part II

Define

$$Y = X^2.$$

(d) Find the range of Y.

(e) Express x as a function of y, and calculate

$$\left|\frac{dx}{dy}\right|.$$

(f) Using the change-of-variables formula

$$p_Y(y) = p_X(x)\left|\frac{dx}{dy}\right|,$$

derive $p_Y(y)$.

(g) Verify that your resulting density integrates to one.

(h) Explain briefly why a probability density cannot simply be transformed as an ordinary function $p_X(g(y))$. What role does the Jacobian term play?

Question 3.

A discrete random variable X can take eight states

$$\{a, b, c, d, e, f, g, h\}$$

with probabilities

$$\left(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{64}, \frac{1}{64}, \frac{1}{64}, \frac{1}{64}\right).$$

(a) For each state, calculate its information content

$$h(x) = -\log_2 p(x).$$

Which observations carry the most information? Which carry the least? Explain the relationship between probability and information.

(b) Calculate the entropy

$$H[X] = -\sum_x p(x) \log_2 p(x).$$

Explain why entropy is an *average amount of information*, rather than the information associated with one particular outcome.

(c) Consider the binary code

X	a	b	c	d	e	f	g	h
Code	0	10	110	1110	111100	111101	111110	111111

Calculate the expected code length.

(d) Compare the expected code length in part (c) with $H[X]$.

(e) Suppose instead that all eight states were equally likely. Calculate the entropy in this case and compare it with the entropy obtained above. Explain intuitively why the uniform distribution has greater entropy.

(f) Explain the statement:

> "More probable states should receive shorter codewords."

Your answer should connect probability, information content, entropy, and average code length.

Question 4.

Suppose $N = 12$ distinguishable objects are distributed among three bins.

Consider two macrostates:

$$A = (10, 1, 1)$$

and

$$B = (4, 4, 4).$$

For occupation numbers n_1, n_2, n_3, define the multiplicity

$$W = \frac{N!}{n_1! n_2! n_3!}.$$

(a) Calculate W_A and W_B. Which macrostate corresponds to more possible microscopic arrangements?

(b) For each macrostate, calculate

$$H_N = \frac{1}{N} \ln W.$$

Which has the larger value?

(c) Define

$$p_i = \frac{n_i}{N}.$$

Calculate

$$H(p) = -\sum_{i=1}^{3} p_i \ln p_i$$

for both macrostates.

(d) Explain why

$$\frac{1}{N} \ln W$$

approaches

$$-\sum_{i} p_i \ln p_i$$

as N becomes large.

You may use Stirling's approximation

$$\ln N! \approx N \ln N - N.$$

(e) Among all probability distributions

$$(p_1, p_2, p_3), \qquad p_1 + p_2 + p_3 = 1,$$

which distribution would you expect to have the largest entropy? Explain your answer conceptually.

(f) Now consider a continuous random variable with a fixed mean μ and variance σ^2.

The Gaussian distribution has the largest differential entropy among all distributions satisfying these constraints. Explain what the *maximum-entropy principle* means in this setting. In particular, explain why the Gaussian can be viewed as making the fewest additional assumptions once the mean and variance are specified.

Question 5.

This question connects information theory with machine learning.

Part I

Suppose the true distribution is

$$p = (0.7, 0.2, 0.1),$$

and two models give

$$q_1 = (0.6, 0.25, 0.15), \qquad q_2 = (0.4, 0.3, 0.3).$$

Use natural logarithms in this part.

(a) Calculate the entropy

$$H(p) = -\sum_x p(x) \ln p(x).$$

(b) Calculate the cross-entropies

$$H(p, q_1) = -\sum_x p(x) \ln q_1(x),$$

and

$$H(p, q_2) = -\sum_x p(x) \ln q_2(x).$$

(c) Calculate

$$\mathrm{KL}(p\|q_1) \qquad \text{and} \qquad \mathrm{KL}(p\|q_2),$$

where

$$\mathrm{KL}(p\|q) = \sum_x p(x) \ln \frac{p(x)}{q(x)}.$$

Verify numerically that

$$H(p, q) = H(p) + \mathrm{KL}(p\|q).$$

(d) Explain why minimizing cross-entropy with respect to q is equivalent to minimizing

$$\mathrm{KL}(p\|q)$$

when p is fixed.

(e) Explain the connection between this result and maximum-likelihood training of a probabilistic machine-learning model.

Part II

Suppose two binary variables have joint distribution

	$Y = 0$	$Y = 1$
$X = 0$	0.40	0.10
$X = 1$	0.10	0.40

.

Use logarithms to base 2 in this part.

(f) Find the marginal distributions $p(X)$ and $p(Y)$.

(g) Calculate

$$H[Y].$$

(h) Find

$$p(Y \mid X = 0) \qquad \text{and} \qquad p(Y \mid X = 1),$$

and hence calculate

$$H[Y \mid X].$$

(i) Calculate the mutual information

$$I(X;Y) = H[Y] - H[Y \mid X].$$

(j) Explain the meaning of $I(X;Y)$ in words. Your explanation should use the idea

$$\boxed{\text{information gained} = \text{uncertainty before} - \text{uncertainty after}}.$$

(k) What would $I(X;Y)$ be if X and Y were independent? Explain your answer without performing a detailed calculation.